Component Sizes — gem5 PNM Simulation

This document catalogs the size of every configurable component in the baseline and PNM gem5 simulations: the dataset, RocksDB internal structures, CPU caches, DRAM, and the PNM hardware device. All numbers are drawn from `configs/yonsei/run_baseline.py`, `configs/yonsei/run_pnm.py`, and the `m5out/baseline_v3/` / `m5out/pnm_v3/` simulation outputs (25,000-key workload).

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1. Dataset Dimensions

Parameter	Value	Bytes	Note
Key count (`--num`)	25,000	—	Unique random keys inserted by fillrandom
Value size (`--value_size`)	1,024 B	1,024	Fixed per value; no compression (`--compression_type=none`)
Key size (db_bench default)	16 B	16	Fixed-width key encoding for fillrandom
Raw key-value payload	~24.8 MiB	26,000,000	25,000 × (16 + 1,024) = 26.0 MB; no compression applied
Internal RocksDB overhead per key	~16–24 B	—	Sequence number (8 B), value type, block/index overhead
Estimated total on-disk size (uncompressed)	~26.4 MiB	~27,600,000	25,000 × (16 + 1,024 + ~20 overhead); confirmed by final SST sizes in Section 5
Seed (`--seed`)	42	—	Deterministic key generation for reproducibility

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2. RocksDB Internal Component Sizes

Component	Flag	Size	Bytes	Purpose
Memtable (single write buffer)	`--write_buffer_size`	1 MiB	1,048,576	In-memory write buffer; flushes to L0 SST when full
Max active memtables	`--max_write_buffer_number`	2	—	Up to 2 MiB of in-flight write data before stalling
Max in-memory write data	(derived)	2 MiB	2,097,152	2 × write_buffer_size; write stall if both full
L0 SST file size	(derived from memtable)	~1 MiB	~1,048,576	Each memtable flush produces one ~1 MiB L0 file; confirmed by compaction src sizes
L0 compaction trigger	`--level0_file_num_compaction_trigger`	2 files	~2 MiB	Compaction fires when L0 has ≥ 2 files (~2 MiB of unflushed L0 data)
L1 maximum size	`--max_bytes_for_level_base`	16 MiB	16,777,216	Total SST data allowed in Level 1; triggers L1→L2 compaction if exceeded
SST block size	`--block_size`	4 KiB	4,096	Unit of read from DRAM; one cache miss = one 4 KiB fetch

Block cache	`--cache_size`	8 MiB	8,388,608	OS-side LRU cache for decompressed SST blocks; shared across all SST reads
Block cache capacity in blocks	(derived)	2,048 blocks	—	8 MiB ÷ 4 KiB = 2,048 blocks simultaneously cached
Compaction style	`--compaction_style=0`	Leveled	—	L0→L1 merges; all compaction jobs in this research are L0→L1

Keys per memtable (estimated): 1,048,576 B ÷ (16 + 1,024 + 20) B = ~985 keys per flush. With 25,000 keys total: ~25 L0 flushes, triggering ~12–13 initial compaction events before cascading merges consolidate into the final L1 state.

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3. Hardware Component Sizes

CPU Caches (per core, both baseline and PNM)

Component	Size	Bytes	Config flag
L1D (data cache)	32 KiB	32,768	`l1d_size="32KiB"`
L1I (instruction cache)	32 KiB	32,768	`l1i_size="32KiB"`
L2 (unified, private)	512 KiB	524,288	`l2_size="512KiB"`
L1D capacity in 4 KiB blocks	8 blocks	—	32 KiB ÷ 4 KiB; only 8 SST blocks fit in L1D simultaneously
L2 capacity in 4 KiB blocks	128 blocks	—	512 KiB ÷ 4 KiB; L2 holds 128 hot SST blocks before evicting to DRAM
Total cache per core (L1D+L1I+L2)	576 KiB	589,824	—
Total cache — baseline (1 core)	576 KiB	589,824	—
Total cache — PNM (2 cores)	1,152 KiB	1,179,648	Separate private caches; no shared L3
CPU clock	3 GHz	—	`clk_freq="3GHz"`
Clock period	0.333 ns	—	1 cycle = 333 ps

DRAM

Component	Value	Note
Type	Dual-channel DDR4-2400	`DualChannelDDR4_2400`
Total capacity	3 GiB	3,221,225,472 bytes — `size="3GiB"`
Channels	2 (mem_ctrl0, mem_ctrl1)	Traffic split evenly across both channels
Capacity per channel	1.5 GiB	1,610,612,736 bytes
Data bus width per channel	64-bit (8 bytes)	Standard DDR4 channel width
Effective transfer size per	64 bytes	8-beat burst × 8 bytes; one cache-line-sized transfer

burst		
Simulated peak BW (both channels)	~36–38 GiB/s	DDR4-2400 × 2 channels theoretical; actual avg RdBW ~123–139 MiB/s in simulation

PNM Device (run_pnm.py only)

Component	Value	Note
MMIO base address	`0xD0000000`	32-bit physical address; mapped via `/dev/pnm`
MMIO region size	256 bytes (0x100)	4 registers: src_bytes, dst_bytes, cmd, status
Process latency	500 ns	AxDIMM-style: DDR4 round-trip (≈80 ns) + buffer-chip decode overhead
Internal bandwidth	25 GiB/s	Rank-parallel internal DIMM bandwidth model
MMIO register: src_bytes	64-bit, offset 0x00	Host writes compaction input file size before CMD_SUBMIT
MMIO register: dst_bytes	64-bit, offset 0x08	Host writes estimated output size before CMD_SUBMIT
MMIO register: cmd	32-bit, offset 0x10	CMD_RESET=2, CMD_SUBMIT=1
MMIO register: status	32-bit, offset 0x14	DONE=bit 1, ERR=bit 2; polled by pnm_compaction_unit after job
Socket path (IPC)	`/tmp/pnm_compaction.sock`	UNIX domain socket between rocksdb_pnm and pnm_compaction_unit
Secondary DB path	`/tmp/pnm_secondary`	Persistent secondary DB for incremental deduplication between compaction jobs

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4. Complete Size Hierarchy

All components ordered from smallest to largest, showing how each fits relative to the others.

Component	Size	Bytes	× larger than L1D
— CPU Hardware —			
L1D cache (per core)	32 KiB	32,768	1× (reference)
L1I cache (per core)	32 KiB	32,768	1×
L2 cache (per core)	512 KiB	524,288	16×
— RocksDB Application —			
SST block size	4 KiB	4,096	0.125× (fits 8 blocks in L1D)
Memtable (single)	1 MiB	1,048,576	32×
L0 SST file (~1 memtable flush)	~1 MiB	~1,048,576	~32×
Max in-flight memtables (2×)	2 MiB	2,097,152	64×
Block cache	8 MiB	8,388,608	256×
Raw KV dataset (25k keys)	~24.8 MiB	26,000,000	793×
L1 max size (config)	16 MiB	16,777,216	512×
— Actual SST State after fillrandom (v3, from simulation) —			

PNM: final L1 SST (last compaction output)	9.1 MiB	9,551,057	291×
Baseline: final L1 SST (last compaction output)	15.8 MiB	16,575,904	506×
— DRAM Traffic (v3 simulation totals) —			
PNM: total bytes written to DRAM	368.5 MB	386,347,008	11,793×
Baseline: total bytes written to DRAM	394.8 MB	413,958,144	12,634×
PNM: total bytes read from DRAM	905.5 MB	949,583,872	28,980×
Baseline: total bytes read from DRAM	1,028.6 MB	1,078,683,648	32,913×
— DRAM Capacity —			
DRAM total	3 GiB	3,221,225,472	98,304×

5. Actual SST File Sizes — From Simulation (v3, 25k keys)

Each compaction job's input and output file sizes, showing how the on-disk data evolves through the compaction sequence. All sizes are in bytes.

Baseline v3 (5 sequential, blocking jobs)

Job	files_in	src size (B)	src size (MiB)	dst size (B)	dst size (MiB)	compression ratio
job=4	2	1,962,302	1.87	1,929,363	1.84	0.98
job=9	4	4,863,354	4.64	4,540,460	4.33	0.93
job=16	7	10,423,228	9.94	9,009,458	8.59	0.86
job=30	14	21,778,286	20.77	15,814,192	15.08	0.73
job=32	3	17,774,374	16.95	16,575,904	15.81	0.93
Final L1 state	—	—	—	**16,575,904**	**15.81**	—

The final L1 SST (15.81 MiB) is nearly at the configured 16 MiB limit. Only 6.7% of the dataset's potential deduplication is achieved by standard RocksDB compaction (8,441 keys dropped, or ~34% of the 25,000 writes represent superseded keys).

PNM v3 (5 concurrent, MMIO-offloaded jobs)

files_in	src size (B)	src size (MiB)	dst size (B)	dst size (MiB)	compression ratio	vs Baseline
2	1,951,045	1.86	1,112,870	1.06	0.57	**−42%**
4	4,023,138	3.84	2,616,400	2.49	0.65	**−42%**
7	8,450,872	8.06	5,190,150	4.95	0.61	**−42%**
13	16,879,284	16.10	8,869,379	8.46	0.53	**−44%**
4	11,750,351	11.21	9,551,057	9.11	0.81	**−42%**
Final L1 state	—	—	**9,551,057**	**9.11**	—	**−42% vs baseline**

PNM's secondary-DB deduplication achieves a consistent 42–44% size reduction per job. The final L1 SST (9.11 MiB) is 42% smaller than baseline's (15.81 MiB).

6. Working Set vs Cache Capacity

The critical size relationships that determine whether data is served from cache or DRAM.

Relationship	Ratio	Implication
— Block Cache (8 MiB) vs SST Working Set —		
Raw dataset / block cache	**24.8 MiB / 8 MiB = 3.1×**	Dataset is 3× the block cache — significant cache eviction during readrandom
Baseline final L1 / block cache	**15.81 MiB / 8 MiB = 1.97×**	Final SST state is ~2× the cache — about half of all blocks must be re-fetched from DRAM on each readrandom pass
PNM final L1 / block cache	**9.11 MiB / 8 MiB = 1.14×**	PNM's denser SSTs bring the working set within 14% of fitting entirely in the block cache — far fewer cold-cache misses during readrandom
— L2 Cache (512 KiB per core) vs Hot Working Set —		
L2 capacity in SST blocks	128 blocks × 4 KiB = 512 KiB	L2 can hold 128 recently accessed SST blocks; the most recently touched data fits before evicting to DRAM
L2 / memtable size	**512 KiB / 1 MiB = 0.5×**	A single memtable is 2× larger than L2 — a memtable flush necessarily evicts all prior L2 content
Baseline L2 miss rate (v3)	**7.68%**	Compaction SST scans continuously evict write-path data from L2 → high miss rate
PNM L2 miss rate CPU0 (v3)	**5.85%**	No compaction evictions from CPU0's L2 → 24% lower miss rate, 64% fewer absolute misses
— L1D Cache (32 KiB per core) vs Data Access Patterns —		
L1D capacity in SST blocks	8 blocks × 4 KiB = 32 KiB	Only 8 SST blocks fit in L1D — SST access has essentially zero L1D reuse between different keys
Baseline L1D miss rate (v3)	**1.977%**	High absolute miss count (20.6M) — compaction scan data pollutes the write-path working set
PNM L1D miss rate CPU0 (v3)	**2.013%**	Rate barely changes but absolute misses drop 45.7% — fewer total memory accesses without compaction work
— DRAM (3 GiB) vs Everything —		
DRAM / raw dataset	**3 GiB / 24.8 MiB = 124×**	Dataset easily fits in DRAM with no pressure; RocksDB's tiered structure (L0/L1) is the bottleneck, not total DRAM
DRAM / block cache	3 GiB / 8 MiB = 384×	DRAM capacity far exceeds the block cache — DRAM latency, not capacity, is the bottleneck

7. DRAM Traffic Volumes (v3 simulation, 25k keys)

Metric	Baseline v3	PNM v3	Change
Total read requests (both channels)	16,071,757	14,148,149	**−12.0%**
Total bytes read (both channels)	1,028.6 MB	905.5 MB	**−12.0%**
Total bytes written (both channels)	394.8 MB	368.5 MB	**−6.7%**
Read requests per key	642	566	**−11.9%**

Bytes read per key	41.1 KiB	36.2 KiB	**−12.0%**
Avg read bandwidth ch0 (simulated)	69.59 MiB/s	61.95 MiB/s	−11.0%
Avg read bandwidth ch1 (simulated)	69.47 MiB/s	61.62 MiB/s	−11.3%
Simulation duration	7.381 s	7.314 s	**−0.9%**

Bytes read per key (41.1 KiB baseline): each of the 25,000 keys generated about 10 DRAM reads of 4 KiB across the fill + read + compaction phases. This is a measure of total read amplification — the ratio of DRAM traffic to raw data size. PNM reduces it to 36.2 KiB per key (−12%), entirely from smaller SST output files requiring fewer block fetches.

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8. Size Summary

Key takeaways from a size perspective:

#	Observation	Impact
1	Block cache (8 MiB) is **3.1× smaller** than the raw dataset (24.8 MiB)	Cache thrash is inevitable during readrandom — both baseline and PNM cannot fully cache the working set
2	PNM's final SST (9.1 MiB) is **within 14%** of fitting in the block cache; baseline's (15.8 MiB) is **97% over**	PNM dramatically narrows the gap between working set and cache, halving the cold-miss penalty during readrandom
3	L2 (512 KiB) can hold only **128 SST blocks**; a single memtable flush (1 MiB) is **2× L2 capacity**	Every memtable flush evicts the entire L2 unless the write path's hot data is protected — baseline cannot protect it from compaction
4	L1D (32 KiB) holds only **8 SST blocks** at a time	SST access is essentially streaming from an L1D perspective — all SST data comes from L2 or DRAM
5	The PNM MMIO region is **256 bytes** (0x100) — 4 registers	The hardware interface is deliberately minimal; the heavy I/O flows through the DRAM channels, not the MMIO bus
6	DRAM capacity (3 GiB) is **124× the dataset size**	DRAM capacity is not the constraint in these simulations — DRAM bandwidth and latency are

All simulation data from m5out/baseline_v3/ and m5out/pnm_v3/ — gem5 full-system x86, db_bench fillrandom,readrandom,waitforcompaction num=25000 value_size=1024 seed=42 cache_size=8388608 block_size=4096 write_buffer_size=1048576.